

Force

****In physics, a force is a push or pull acting upon an object as a result of the object's interaction with another object.****

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- ****Vector Quantity**** – Forces have both magnitude (strength) and direction.
- ****SI Unit**** – The standard metric unit for force is the ****Newton (N)****.
- ***One Newton is the force required to accelerate a 1 kg mass at a rate of 1 m s⁻².***
- ****Formula**** – Newton's Second Law of Motion:

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$$\boxed{F = ma}$$

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where

- (F) is force,
- (m) is mass, and
- (a) is acceleration.

Common Types of Forces

Forces are generally categorized into two main groups:

1. Contact Forces

Applied through physical contact between objects.

- **Friction** – resistance to relative motion.
- **Tension** – pulling force transmitted through a string or cable.
- **Air resistance** – drag force opposing motion through a fluid.
- **Normal force** – perpendicular reaction force from a surface.
- **Applied push/pull** – direct mechanical forces.

2. Non-Contact (Field) Forces

Act over a distance without physical contact.

- **Gravitational** – attraction between masses.
- **Electromagnetic** – attraction/repulsion between charged particles or magnetic fields.
- **Strong nuclear** – binding force between nucleons in an atomic nucleus.
- **Weak nuclear** – responsible for radioactive decay.

Quick Reference

Concept	Symbol	Units	Typical Value
Force	(F)	N (newtons)	$1\text{ N} = 1\text{ kg}\cdot\text{m s}^{-2}$
Mass	(m)	kg	1 kg
Acceleration	(a)	m s^{-2}	1 m s^{-2}

Remember:

- Forces are vectors; always indicate direction.

- Use Newton's Second Law to calculate unknowns when mass and acceleration are known.
- Distinguish between contact and non-contact forces when analyzing a system.